

How much energy does a data center use?

Posted by u/Kim-2000 • 0 points • 58 comments

I have seen reports that say a single server rack consumes between 10-15kW of energy, meanwhile an GPU server rack uses between 40-60kW. And the average data center has around 10k racks. Can it really be that a data center uses upward of 100MW?

Comments

u/Dull-Communication82 • 1 points • 2025-09-22 05:55 UTC

You can check out energy related stats here: [<https://brightlio.com/data-center-stats/>]
(<https://brightlio.com/data-center-stats/>)

u/Pretend-Extreme7540 • 1 points • 2025-09-16 12:59 UTC

The colossus data center is expected to consume 150 MW.

[[https://en.wikipedia.org/wiki/Colossus_\(supercomputer\)](https://en.wikipedia.org/wiki/Colossus_(supercomputer))]
([https://en.wikipedia.org/wiki/Colossus_\(supercomputer\)](https://en.wikipedia.org/wiki/Colossus_(supercomputer)))

Apparently some data centers are planned that are supposed to require 1GW+

u/Sad-Juice3667 • 1 points • 2025-08-16 17:26 UTC

yoooo i need to cite this for a essay where can i find this online

u/[deleted] • 1 points • 2025-04-21 22:04 UTC

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u/[deleted] • 1 points • 2025-01-28 16:24 UTC

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u/[deleted] • 1 points • 2024-12-29 21:03 UTC

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u/[deleted] • 1 points • 2024-11-19 16:19 UTC

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u/Intrepid-Refuse-9901 • 1 points • 2024-11-19 11:55 UTC

You're correct; large data centers can consume massive amounts of energy, often reaching 100MW or more, depending on their size and purpose. GPU-intensive workloads, like AI or rendering, significantly increase power usage. Efficiency measures like advanced cooling systems and renewable energy integration help offset this. It's a big energy footprint, but also critical for our digital infrastructure.

u/SomeDude_91 • 1 points • 2025-06-01 20:29 UTC

That's about 100 MW of electricity being converted to heat. Maybe with water-cooling, that excess heat (lets say, water at 60-80 Celsius) could be used to heat up houses or offices.

u/Intrepid-Refuse-9901 • 1 points • 2024-11-05 07:13 UTC

Yes, data centers can consume massive amounts of power—some even exceed 100MW depending on their size and setup. Power usage varies with the type of equipment and the facility's cooling and redundancy systems. High-density GPU racks, for instance, demand much more energy due to their intense processing loads, especially in AI and machine learning environments. That's why energy efficiency and cooling innovations are key priorities for data centers to manage both costs and environmental impact.

u/OneLifeguard8422 • 1 points • 2024-09-26 00:58 UTC

They can certainly consume over 100MW. It really depends on the size of the data center and the configuration. Small-size data centers have a power consumption of up to 20MW, while mid-size facilities can consume between 50-70MW and the larger ones can be from 100MW to 250MW.

The largest one to date has over 500MW of power generation capacity. majority of them used to consume power from the grid, but with the grid reaching its limits, the other solution is to build either solar/wind farms or use power generation packages to support the power demand.

See this for the solar farm: [<https://www.solaralliance.com/commercial-solar-arrays/data-centers/>]
(<https://www.solaralliance.com/commercial-solar-arrays/data-centers/>)

Also, see this for the power generation: <https://metispower.com/home/industries/data-centers-power-solutions/>

u/jeff61813 • 2 points • 2024-08-16 00:33 UTC

When you build a data center usually in the industry people just say how many megawatts it's going to consume, Like oh that's a really small facility. It's 50 megawatts, or that's a very large facility That's 200 megawatts